

Unified Multi-Environment Cyber Threat Detection & Response Platform

Autonomous Telemetry Ingestion • MITRE ATT&CK Kill-Chain • ML Anomaly Engine • Google Gemini 2.0 Flash • SHA-256 Audit Ledger

DEPARTMENT OF INFORMATION TECHNOLOGY
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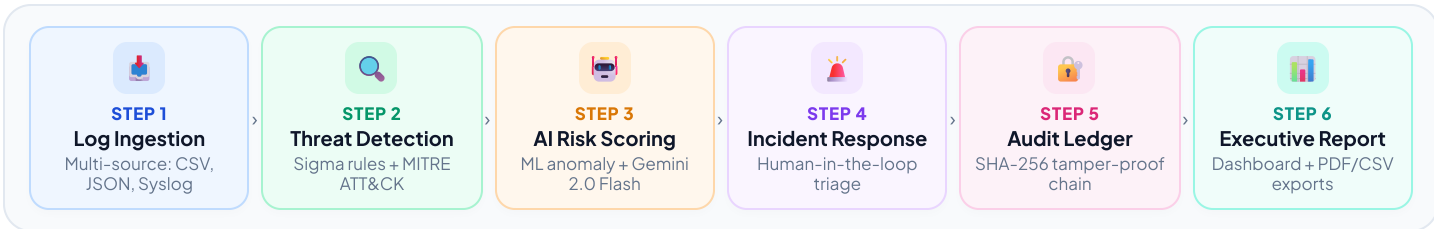
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OBJECTIVE

Build a **real-time, autonomous Security Operations Center (SOC)** platform that unifies log ingestion, MITRE ATT&CK-mapped threat detection, ML-based anomaly scoring, human-in-the-loop incident response, and cryptographic tamper-proof audit logging – with a **Google Gemini 2.0 Flash (Latest Release) AI Threat Analyst** chatbot for interactive threat intelligence.

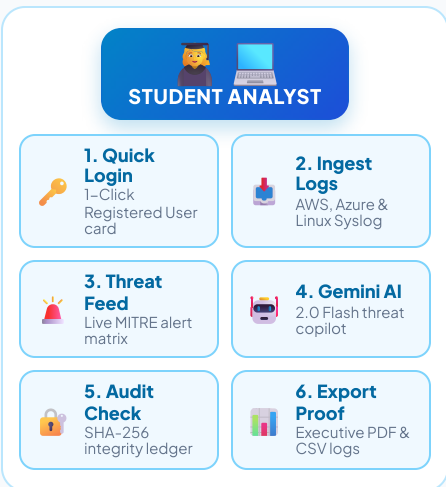
SYSTEM PROJECT WORKFLOW (PIPELINE FLOW)



Key Features

- ✓ **Real-time Threat Detection** — Sigma rules on live ingestion stream
- ✓ **MITRE ATT&CK Kill-Chain** — 7-stage tactic/technique mapping
- ✓ **ML Anomaly Engine** — Composite risk score 0-100
- ✓ **Gemini 2.0 Flash AI Analyst** — Multimodal LLM chatbot for threat Q&A
- ✓ **Human-in-the-Loop** — Analyst approve/escalate/close workflows
- ✓ **SHA-256 Audit Chain** — Cryptographic tamper-evident proof
- ✓ **Registered User Login** — Auto-recognizes registered operators

Student View (Platform Experience)



Technologies Used

- FRONTEND**
 - React 18
 - TypeScript
 - Vite 5
 - Tailwind CSS
- BACKEND**
 - FastAPI
 - SQLite / DB
 - Redis Queue
 - JWT Auth
- AI & INTELLIGENCE**
 - Gemini 2.0 Flash (Latest)
 - MITRE ATT&CK v14
 - AbuseIPDB API
 - SHA-256 Crypto
- DEPLOYMENT**
 - Vercel (Prod)
 - GitHub Actions
 - Docker
 - WebSocket

❌ 2026 Real Breach (Legacy Disconnected SOC)

Attackers (UNC3886) moved laterally across AWS & Azure hypervisors undetected for 180+ days • Deleted Linux audit logs to hide tracks • Siloed SIEM tools caused alert blindness • Forensics took 4 months.

✅ Present Unified Platform Defense (With AI)

ECS 1.12 normalizer flags lateral movement in <5 min • MITRE T1078 & T1021 auto-mapped • SHA-256 chain detects log tampering instantly • Gemini 2.0 Flash gives real-time containment commands in 4 hours.

Live Project Links



unified-threat-
platform.vercel.app



github.com/Pravallika2025/unified-
threat-platform



Gemini
AI:

Gemini
2.0
Flash
Free
Key →



admin@threatplatform
Login: / Admin@12345